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Experiment M3-bis2
Interfereneces in the Quincke tube

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1 Introduction

The Quincke tube (figure 1) is a classic experimental apparatus used to study acoustic wave interference. By splitting a sound wave into two paths of different lengths, the device allows for the observation of constructive and destructive interference patterns at a detector. This phenomenon occurs due to the phase difference introduced by the path length variation, providing a direct method to analyze the properties of sound propagation [1].

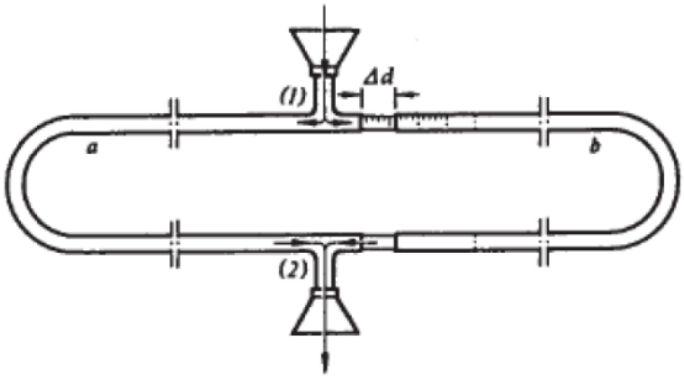


Figure 1: Quincke tube [2].

The objective of this study is to determine the experimental speed of sound and characterize the physical geometry of the tube. To achieve this, two procedures were performed: a resonance frequency sweep (Experiment A) and a displacement sweep at a constant frequency (Experiment B) [2]. The experimental results are compared with theoretical models to validate the precision of the measurements and the interference theory.

2 Results and discussion

2.1 Experimental measurements

Table 3 and 4 present experimental data collected in the laboratory. For the first experience, the objective was to identify the maximal and minimal values of the pressure wave, measuring their voltage V and frequency f. In the second experiment, on the other hand, the main purpose was to measure the amplitude a of the wave when changing the distance s of the Quincke tube.

In addition to the experimental dataset, the uncertainties of each magnitude are included in both tables. *The oscilloscope graphics requested in the manual were not included in this report due to the indications of our professors.

2.2 Resonance frequency in experiment A

The graph 2 presents the experimental results for experiment A. By plotting the resonant frequency f_n on the against the relative order number n, we can observe the behavior of the air inside the Quincke tube. The data points follow a clear linear trend, which is consistent with the theoretical model $f_n = n \cdot (c/2a)$.

The non-zero y-intercept observed in the plot suggests that the order n used is a relative index rather than an absolute one, which is expected since the frequency sweep began at a specific starting point of 1300 Hz, as the manual stated.

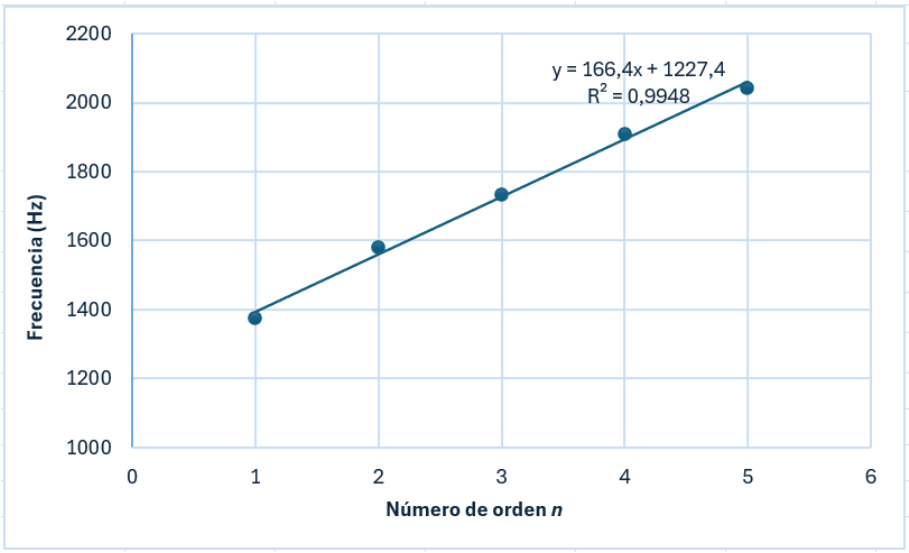


Figure 2: Resonance frequency over order n.

The linear regression applied to these points is presented in Table 1. The high coefficient of determination confirms a strong correlation between the variables and validates the use of the theoretical formula for this analysis.

m **(Hz)**	d **(Hz)**	c/a **(Hz)**
166 ± 7	1227 ± 23	$332,8 \pm 9,8$

Table 1: Values of the fit m and d, and c/a, with their respective errors.

The relation c/a was determined using the following equation.

$$f_n = n\frac{c}{2a}, \tag{1}$$

where the slope m represents the coefficient $c/2a$. Therefore, we conclude that the characteristic ratio of the Quicke tube is

$$\boxed{c/a = 332,8 \pm 9,8 \quad Hz}$$

2.3 Graphic representation of experiment B

Figure 3 illustrates the results from Experiment B, showing the variation in sound amplitude as a function of the mobile arm displacement s. The data reveals a oscillatory pattern with three distinct maximums and minimums, which confirms constructive and destructive interferences within the Quincke tube.

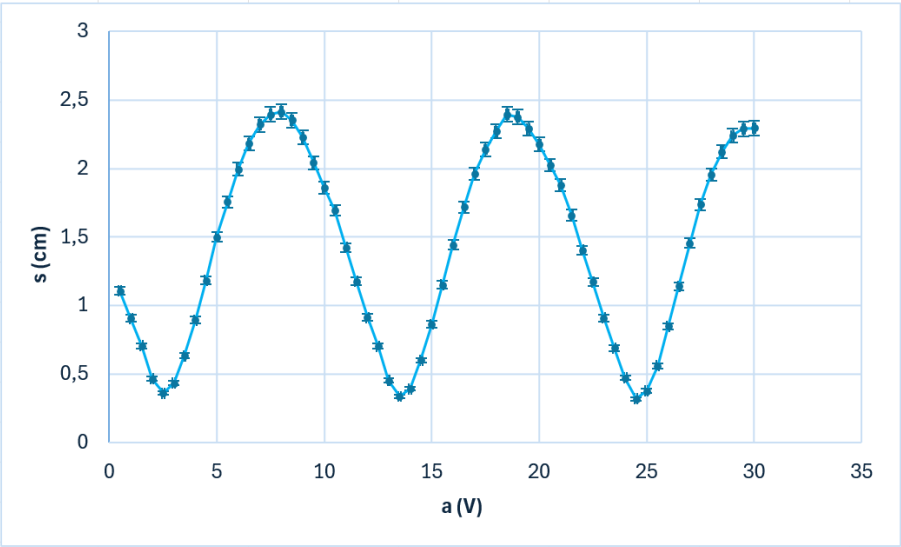


Figure 3: Amplitude a over displacement s of the Quincke tube in experiment B.

2.4 Determination of the Speed of Sound

Based on the experimental data from Table 1, the positions of the interference maxima and minima were identified. Then, a linear regression was performed by plotting the displacement s (in meters) against the relative order number n:

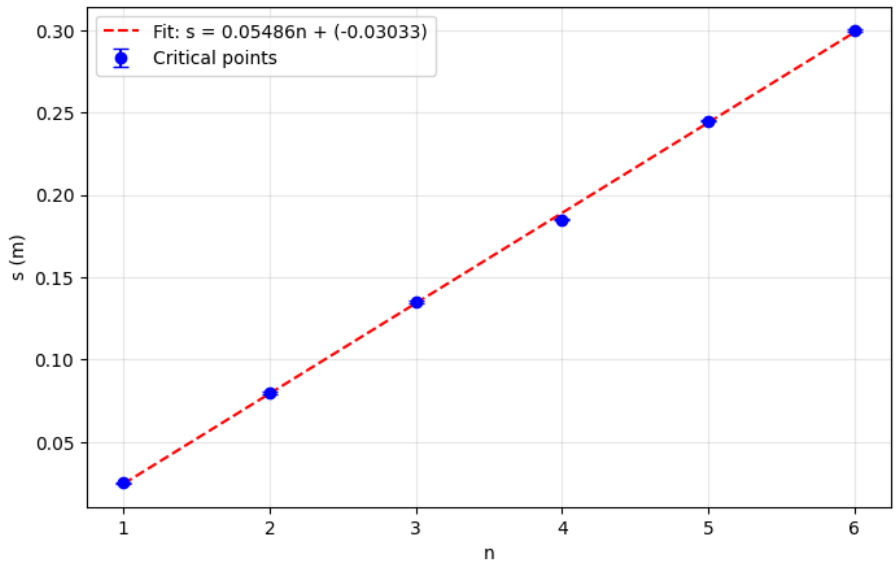


Figure 4: Linear regression performed in order to obtain the C experimental value.

It is important to note that n is treated as a relative index rather than an absolute one, as the measurement begins at an arbitrary point on the Quincke tube's scale. This choice is reflected in the non-zero y-intercept b.

The theoretical relationship between the displacement and the order of interference is given by [2]:

$$s = n \cdot \left(\frac{c}{4f} \right) + b \tag{2}$$

where c is the speed of sound and f is the fixed frequency of the source ($1579,0 \pm 0,58$ Hz). The results of the linear fit are summarized in Table 2.

Parameter	**Value**
Slope (m)	$0,05486 \pm 0,00054$ m
Intercept (b)	$-0,03033$ m
Correlation coefficient (R^2)	$0,99961$

Table 2: Linear fit parameters for Experiment B.

The speed of sound c was calculated from the slope as $c = 4 \cdot f \cdot m$, obtaining that way the following experimental value:

$$c_{exp} = 346,48 \pm 3,42 \text{ m/s} \tag{3}$$

To validate this result, it was compared with the theoretical value at the laboratory temperature ($\theta = 25,5°\text{C}$), calculated using $c = 331,3\sqrt{1 + \theta/273,15}$:

$$c_{theo} = 346,42 \pm 0,068 \text{ m/s} \tag{4}$$

The relative error between the experimental and theoretical values is $0,02\%$. This extremely low error, combined with an R^2 value very close to unity, confirms the high precision of the measurements and the validity of the interference model in the Quincke tube.

2.5 Determination of the Characteristic Length a

The objective of this section is to determine the characteristic length a of the Quincke tube using the experimental results obtained in the previous sections. From the resonance condition analyzed in section R2, the ratio between the speed of sound and the characteristic length was determined to be:

$$c/a = 332,8 \pm 9,8 \text{ Hz} \tag{5}$$

By utilizing the experimental value of the speed of sound calculated in section R4, $c_{exp} = 346,48 \pm 3,42$ m/s, the characteristic length a can be isolated as follows:

$$a = \frac{c_{exp}}{(c/a)} \tag{6}$$

Substituting the experimental values:

$$a = \frac{346,48}{332,8} \approx 1,041 \pm 0,064 \text{ m} \tag{7}$$

This result represents the effective path length of the fixed arm of the Quincke tube. The consistency between the measurements in Experiment A (frequency sweep) and Experiment B (displacement sweep) allows for a reliable characterization of the experimental setup's geometry.

3 Conclusion

The experimental study of acoustic wave interference in the Quincke tube has successfully demonstrated the principles of constructive and destructive interference. By analyzing the displacement of the mobile arm at a fixed frequency of $1579,0 \pm 0,58$ Hz, it was possible to determine the experimental speed of sound as $c_{exp} = 346,48 \pm 3,42$ m/s. As it was discused, this result shows an excellent agreement with the theoretical value of $346,42 \pm 0,07$ m/s, providing an extremely low relative error of $0,02\%$ an a correlation coefficient (R^2) of 0,99961. For that reason, we can confirm the high precision of the measurements and validate the interference model

$s = n \cdot (c/4f) + b$ used for the analysis.

Furthermore, the characterization of the Quincke tube was completed by integrating the results from both experimental procedures. Using the ratio $c/a = 332,8 \pm 9,8$ Hz obtained from the resonance frequency sweep and the previously calculated speed of sound, the characteristic length of the device was identified as $a \approx 1,041 \pm 0,064$ m. This parameter represents the effective path length of the fixed arm of the tube. The consistency observed between Experiment A and Experiment B allowed for a reliable characterization of the experimental setup's geometry.

In conclusion, the results highlight the effectiveness of the Quincke tube as a high-precision tool for studying wave properties in a gaseous medium. The strong correlation between the experimental data and the theoretical models $f_n = n(c/2a)$ and $s = n(c/4f) + b$ demonstrates that the system follows the predicted laws of acoustic interference and resonance.

4 Appendix

4.1 Graphics and fittings

4.2 Data tables

n	V_{max} **(V)**	f_{max} **(Hz)**	V_{min} **(V)**	f_{min} **(Hz)**
1	$1,919 \pm 0,044$	$1373,00 \pm 0,58$	$1,045 \pm 0,027$	$1456,00 \pm 0,58$
2	$2,448 \pm 0,055$	$1579,00 \pm 0,58$	$0,895 \pm 0,024$	$1666,00 \pm 0,58$
3	$1,627 \pm 0,039$	$1732,00 \pm 0,58$	$0,438 \pm 0,015$	$1840,00 \pm 0,58$
4	$1,216 \pm 0,030$	$1909,00 \pm 0,58$	$0,413 \pm 0,014$	$1989,00 \pm 0,58$
5	$0,803 \pm 0,022$	$2040,00 \pm 0,58$	$0,2732 \pm 0,0065$	$2210,00 \pm 0,58$

Table 3: Frequency and voltage for the n maximum and minimum, with their corresponding uncertainties.

s **(cm)**	a **(V)**
0,5 ± 0,058	1,109 ± 0,028
1,0 ± 0,058	0,908 ± 0,024
1,5 ± 0,058	0,708 ± 0,020
2,0 ± 0,058	0,467 ± 0,015
2,5 ± 0,058	0,363 ± 0,013
3,0 ± 0,058	0,436 ± 0,015
3,5 ± 0,058	0,635 ± 0,019
4,0 ± 0,058	0,895 ± 0,024
4,5 ± 0,058	1,184 ± 0,030
5,0 ± 0,058	1,502 ± 0,036
5,5 ± 0,058	1,756 ± 0,041
6,0 ± 0,058	1,995 ± 0,046
6,5 ± 0,058	2,183 ± 0,050
7,0 ± 0,058	2,320 ± 0,052
7,5 ± 0,058	2,393 ± 0,054
8,0 ± 0,058	2,413 ± 0,054
8,5 ± 0,058	2,352 ± 0,053
9,0 ± 0,058	2,227 ± 0,051
9,5 ± 0,058	2,043 ± 0,047

s **(cm)**	a **(V)**
10,0 ± 0,058	1,860 ± 0,043
10,5 ± 0,058	1,696 ± 0,040
11,0 ± 0,058	1,423 ± 0,034
11,5 ± 0,058	1,178 ± 0,030
12,0 ± 0,058	0,915 ± 0,024
12,5 ± 0,058	0,705 ± 0,020
13,0 ± 0,058	0,456 ± 0,015
13,5 ± 0,058	0,339 ± 0,013
14,0 ± 0,058	0,395 ± 0,014
14,5 ± 0,058	0,603 ± 0,018
15,0 ± 0,058	0,865 ± 0,023
15,5 ± 0,058	1,153 ± 0,029
16,0 ± 0,058	1,444 ± 0,035
16,5 ± 0,058	1,719 ± 0,040
17,0 ± 0,058	1,963 ± 0,045
17,5 ± 0,058	2,139 ± 0,049
18,0 ± 0,058	2,270 ± 0,051
18,5 ± 0,058	2,393 ± 0,054
19,0 ± 0,058	2,375 ± 0,054
19,5 ± 0,058	2,290 ± 0,052
20,0 ± 0,058	2,175 ± 0,050
20,5 ± 0,058	2,024 ± 0,046
21,0 ± 0,058	1,880 ± 0,044
21,5 ± 0,058	1,660 ± 0,039
22,0 ± 0,058	1,404 ± 0,034
22,5 ± 0,058	1,173 ± 0,029
23,0 ± 0,058	0,911 ± 0,024
23,5 ± 0,058	0,691 ± 0,020
24,0 ± 0,058	0,473 ± 0,015
24,5 ± 0,058	0,322 ± 0,012
25,0 ± 0,058	0,380 ± 0,014
25,5 ± 0,058	0,561 ± 0,017
26,0 ± 0,058	0,850 ± 0,023
26,5 ± 0,058	1,143 ± 0,029
27,0 ± 0,058	1,455 ± 0,035
27,5 ± 0,058	1,739 ± 0,041
28,0 ± 0,058	1,954 ± 0,045
28,5 ± 0,058	2,123 ± 0,048
29,0 ± 0,058	2,238 ± 0,051
29,5 ± 0,058	2,289 ± 0,052
30,0 ± 0,058	2,295 ± 0,052

Table 4: Wave amplitude a and displacement s of the arms of the Quincke tube.

4.3 Uncertainties

4.3.1 General Uncertainties Calculations

As established in measurement uncertainty analysis, Type A uncertainty, denoted as $u_A(x)$, quantifies the statistical variation associated with repeated observations and is calculated using

the equation:

$$u_A(x) = \frac{\sigma(x)}{\sqrt{N}}, \tag{8}$$

where $\sigma(x)$ represents the standard deviation and N the number of observations. On the other hand, Type B uncertainty, $u_B(x)$, is derived from the estimated accuracy of the measuring instrument and is obtained from:

$$u_B(x) = \frac{\delta_x}{\sqrt{12}}, \tag{9}$$

denotes the instrument's resolution ($\delta_V = 0.01$ V for voltage measurements and $\delta_T = 0.001$ s for the photogate period measurements). The combined standard uncertainty, $u_C(x)$, is subsequently expressed as the root sum of squares of these components:

$$u_C(x) = \sqrt{u_A^2(x) + u_B^2(x)}. \tag{10}$$

For derived quantities obtained through functional relationships $y = f(x_1, x_2, \ldots, x_n)$, the uncertainty was propagated using:

$$u_C(y) = \sqrt{\sum_{i=1}^{n} \left(\frac{\partial f}{\partial x_i}\right)^2 u_C(x_i)^2}. \tag{11}$$

Finally, the expanded uncertainty is defined as:

$$U_C(x) = k \cdot u_C(x), \tag{12}$$

where k represents the coverage factor. Although the GUM [3] allows for various approaches depending on the sample size, we have adopted a coverage factor of $k = 2$. This choice is justified by the large volume of data collected (approximately 15000 points per experiment). With such a significant sample size, the distribution of the results effectively converges to a normal distribution. In this context, a factor of $k = 2$ provides a confidence level of approximately 95% [4].

4.3.2 Direct measurements

A 3360 DMM Peaktech multimeter was used for the voltage measurements. The associated error is given by the manufacturer, PeakTech. In this case, we measured voltage in alterning current, therefore, the precission of the measurment was $\pm 1\%\ rdg + 3\ dgt$ [5].

On the other hand, the frequency and displacement s were measured with a function generator of $\delta f = 1$ Hz and a rule of $\delta x = 0,1$ centimeters of resolution. Therefore,

$$U_C(f) = \frac{2}{\sqrt{12}} = 0,58 \quad Hz, \tag{13}$$

$$U_C(s) = \frac{0,2}{\sqrt{12}} = 0,058 \quad cm. \tag{14}$$

4.3.3 Specific Uncertainty Calculations

The combined uncertainties for each indirect magnitude are shown below. To determine the expanded uncertainty, as previously mentioned, the value will simply be multiplied by a coverage factor of $k = 2$.

- **Coefficient** c/a

$$u_C(c/a) = \sqrt{2} \cdot u_C(m) \quad (Hz) \tag{15}$$

- **C experimental value** c_{exper}

$$u_C(c_{exp}) = c \cdot \sqrt{\left(\frac{u_C(f)}{f}\right)^2 + \left(\frac{u_C(m)}{m}\right)^2}(m/s) \tag{16}$$

(by propagating the errors of the frequency $u_C(f)$ and the slope $u_C(m)$)

- **C theorical value** c_{theo}

$$u_C(c_{theo}) = \sqrt{\left(\frac{\partial c_{th}}{\partial \theta}\right)^2 u_C(\theta)^2} = \frac{331,3}{2 \cdot 273,15\sqrt{1 + \frac{\theta}{273,15}}} \cdot u_C(\theta)(m/s) \tag{17}$$

- **Parameter** a

$$u_C(a) = a \cdot \sqrt{\left(\frac{u_C(c_{exp})}{c_{exp}}\right)^2 + \left(\frac{u_C(c/a)}{(c/a)}\right)^2}(m) \tag{18}$$

References

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